Project Pigeon

Interfacing drones with natural language.

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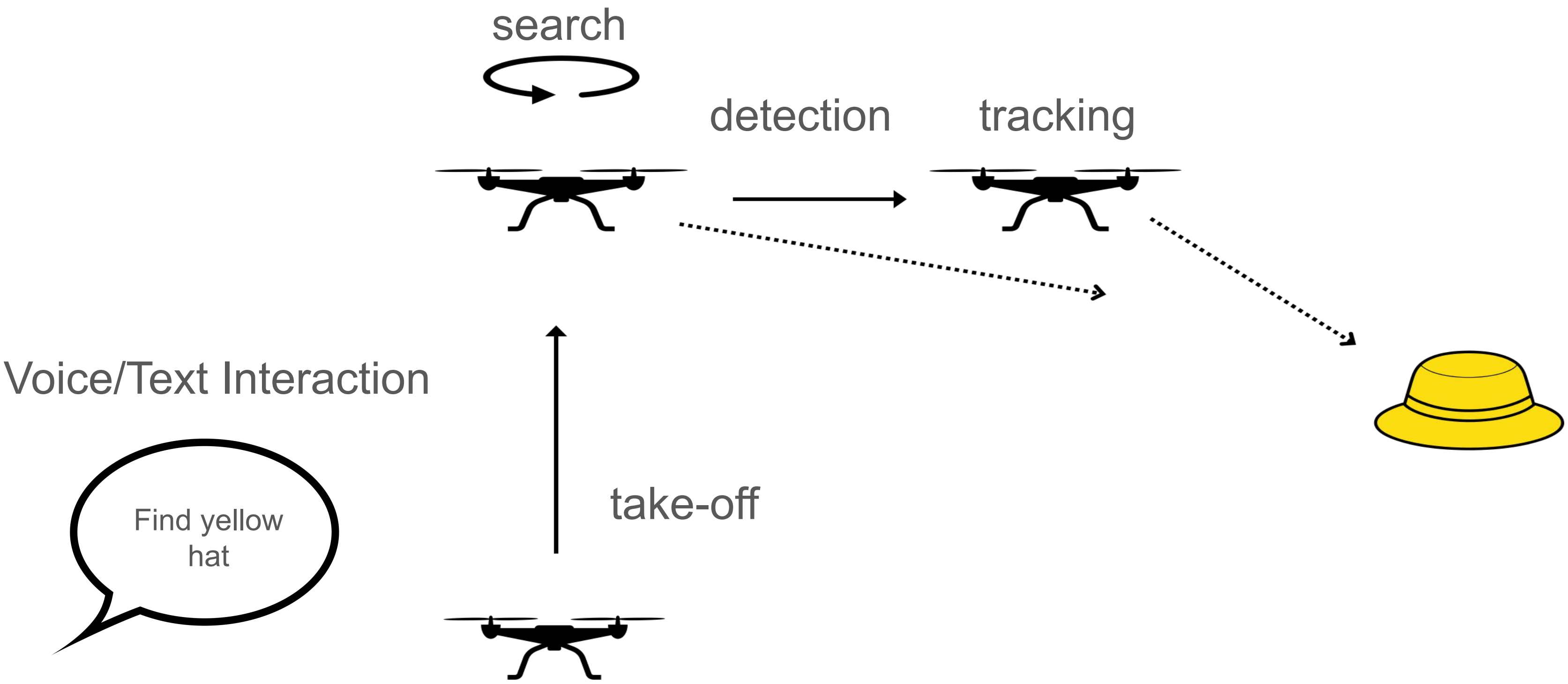
Interface: Natural Language UI, autonomous control



Detection: Open-Vocabulary VLMs

OWL-ViT $\xleftrightarrow{\text{switching}}$ YOLO

OWL-ViT

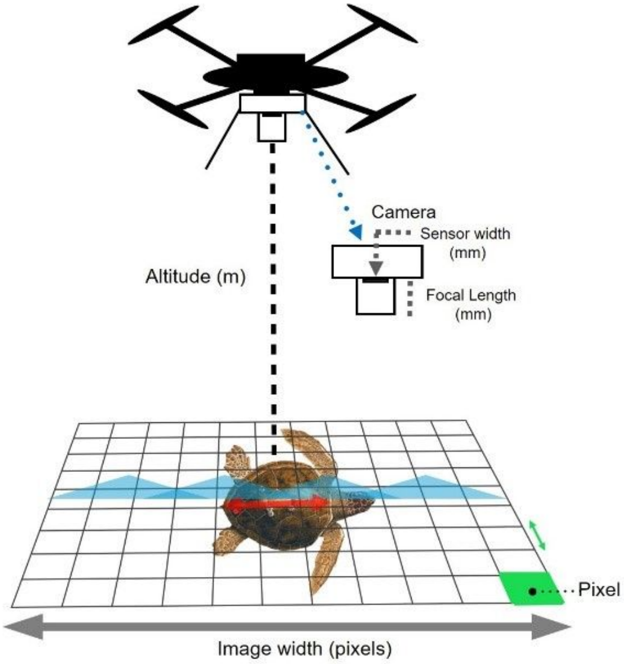
- Open vocabulary detection
- Natural language queries
- Zero-shot capabilities
- 5 FPS

YOLO

- Fast object detection
- Pre-trained classes
- Real-time processing
- 15 FPS

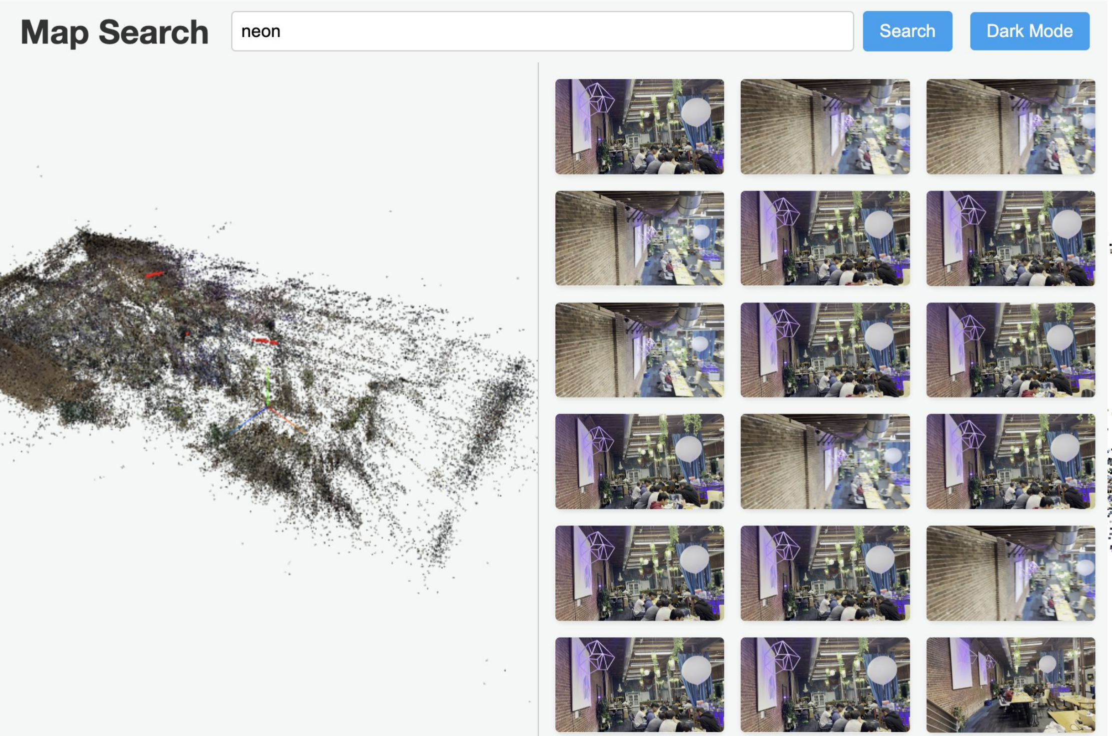
Tracking

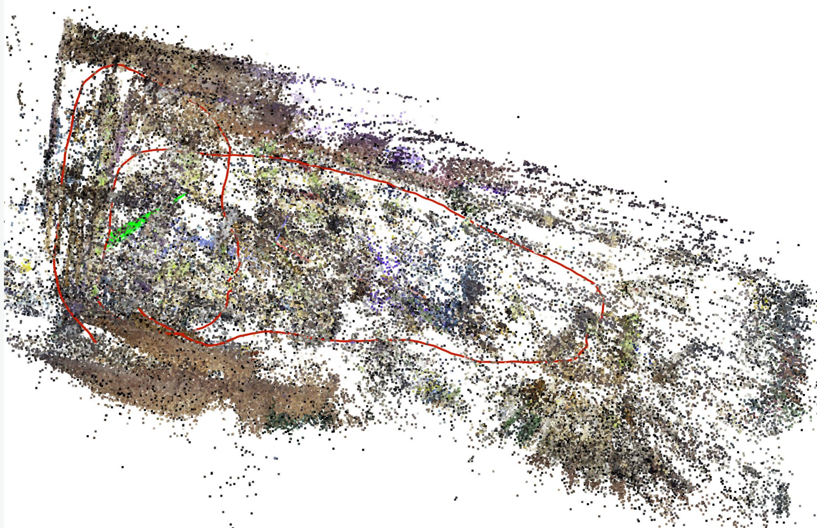
- PID controller to determine control velocities
- Error = (center of target bounding box) - (center of the drone's frame)
- Distance is maintained by setting a bounds for the area of the bounding box, moving back and forth to adjust the size of the box
    - Can use monodepth to maybe compute expected height for input labels



semantic spatial search

- Finds relevant locations in a map
- SLAM based sparse 3D point cloud
- semantic frame search → relevant 3D poses





Video Demo: